

CRT day-3 Test

1) Pattern 5 – Alphabet Triangle

Write a program to print an alphabet triangle pattern. Each row should print alphabets from A up to that row's position.

Test Case 1

Input: 4

Output:

A

AB

ABC

ABCD

Test Case 2

Input: 3

Output:

A

AB

ABC

2) Pattern 12 – Floyd's Triangle

Print Floyd's Triangle using consecutive numbers row by row.

Test Case 1

Input: 4

Output:

1

2 3

4 5 6

7 8 9 10

Test Case 2

Input: 5

Output:

1

2 3
4 5 6
7 8 9 10
11 12 13 14 15

3) GCD of 2 Numbers

Find the Greatest Common Divisor (GCD) of two given numbers.

Test Case 1

Input: 12 18

Output: 6

Test Case 2

Input: 24 36

Output: 12

4) Find Largest Digit

Find the largest digit present in a given number.

Test Case 1

Input: 98341

Output: 9

Test Case 2

Input: 52764

Output: 7

5) Sum of Prime Numbers

Write a program to find the sum of all prime numbers from 1 to N. A prime number has exactly two factors: 1 and itself.

Example:

For N = 10, prime numbers are 2, 3, 5, and 7, so sum = 17.

Test Case 1

Input: 10

Output: 17

Test Case 2

Input: 20

Output: 77

Test Case 3

Input: 5

Output: 10

6) Harshad Number

Write a program to check whether a given number is a Harshad Number. A Harshad Number is divisible by the sum of its digits.

Example:

18 $\rightarrow$ 1 + 8 = 9, and 18 is divisible by 9, so it is a Harshad Number.

Test Case 1

Input: 18

Output: Harshad Number

Test Case 2

Input: 19

Output: Not a Harshad Number

Test Case 3

Input: 21

Output: Harshad Number